

Delving into the Theory of Normal Operators

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Abstract

On finite dimensional spaces, the spectral theorem allows for the decomposition of normal operators into linear combination of orthogonal projections onto the eigenspaces of the operators. The goal of this paper is establish a framework for the spectral theorem for normal operators on spaces of arbitrary dimension and related results thereof. For linear operators on finite dimensional spaces, the spectrum of an operator is the set of eigenvalues for that operator. On infinite dimensional spaces, the spectrum of an operator may contain values that are not eigenvalues. Other particular results of interests include the functional calculus and the spectral mapping theorem. Dealing with operators on infinite dimensional spaces, some of the things we will make use of include measure theory, operator topologies, and properties of C^* -algebras. All the while, we would like for this framework to agree with the finite dimensional case.

1 Introduction

As per the title, we aim to dive into the theory of normal operators on a Hilbert space $\mathcal{H}$. Normal operators possess a litany of applications in various fields. A major one being that of quantum mechanics. In addition, we will define projection-valued measures, which have applications in quantum communication and cryptography [12]. As such, I aim to build a strong understanding of the theory of normal operators. With this, the focus of this paper will be on the theory of normal operators with various examples to promote a strong understanding. As a result, we will forego exploring applications. Normal operators satisfy a rather simple commutation result. But this simple relation has massive results. Moreover, these operators extend far beyond the form they take on in finite dimensional spaces. Nevertheless, matrices serve as a decent starting point for our exploration. In addition, we will also look at how the theory of normal operators holds when dealing with infinite dimensional spaces. In these spaces, traditional linear algebra begins to break down, and so we need to turn to results pertaining to functional analysis to ensure everything remains coherent. This includes notions from measures which take on operators as values, and the various topologies one can define. After going through everything we can, we will briefly mention some next steps.

2 Background information

Starting off our journey into the theory of normal operators, we look at the analytical properties of operator spaces and define measures which take on linear operators as values. Once we get into the theory of normal operators proper, we will look at various examples. Where possible, finite dimensional examples will be used to ensure a concrete understanding. Additionally, throughout this paper, we will assume $0 \notin \mathbb{N}$, and use $\mathbb{N}_0$ when we wish to indicate that the set includes 0.

2.1 Operator Topologies

When we move from finite to infinite dimensional spaces, the topology of the space becomes important. So this is where we will begin our journey.

Definition 2.1. Let $(X, \mathcal{T})$ be a topological space. A net in X is an indexed collection $\{x_k\}_{k \in I} \subseteq X$. This definition is very similar to that of a sequence in a topological. An important distinction to be aware of is that nets may be indexed by an uncountable set. Like sequences, nets carry a similar notion of converge. In this manner, one can think of nets as generalizing of sequences.

Definition 2.2. A *Banach algebra* is a Banach space $\mathcal{A}$ with a multiplication satisfying

$$\|xy\| \leq \|x\| \|y\|$$

for all $x, y \in \mathcal{A}$.

Definition 2.3. A C^* -algebra is a Banach algebra $\mathcal{A}$ with an involution $*$: $\mathcal{A} \rightarrow \mathcal{A}$ such that

- i) $(x^*)^* = x$
- ii) $(xy)^* = y^* x^*$
- iii) $(ax + by)^* = \bar{a}x^* + \bar{b}y^*$
- iv) $\|x^*x\| = \|x\|^2$

for any $x, y \in \mathcal{A}$ and any $a, b \in \mathbb{C}$.

Definition 2.4. A *Hilbert space* is an inner product space $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ so that $\mathcal{H}$ is complete with respect to the metric

$$\begin{aligned} d_{\mathcal{H}}(f, g) &= \sqrt{\langle f - g, f - g \rangle_{\mathcal{H}}} \\ &= \|f - g\|_{\mathcal{H}} \end{aligned}$$

Remark 2.5. For any Hilbert space $\mathcal{H}$, $\mathcal{B}(\mathcal{H})$ is a C^* -algebra with the operator norm topology. As another example, fix some $T \in \mathcal{B}(\mathcal{H})$. Denote by $C^*(T)$ the smallest C^* -subalgebra of $\mathcal{B}(\mathcal{H})$ containing T and T^* . An equivalent definition of $C^*(N)$ is the operator norm closure of polynomials in N and N^* .

Definition 2.6. A *locally convex space* is a topological vector space $\mathcal{V}$ whose topology is defined by a collection of seminorms $\{p_j\}_{j \in I}$ such that

$$\bigcap_{j \in I} \{v \mid p_j(v) = 0, v \in \mathcal{V}\} = \{0\}$$

With this, we are now ready to define the topologies we will need. Recall that for a vector space $\mathcal{A}$, the dual space to $\mathcal{A}$ is the space of linear functionals on $\mathcal{A}$.

Definition 2.7. Let $\mathcal{A}$ be a normed space and let $\mathcal{A}^*$ denote its dual. For $\ell \in \mathcal{A}^*$, define $p_{\ell}: \mathcal{A} \rightarrow \mathbb{R}$ by $p_{\ell}(x) = |\ell(x)|$. Notice that p_{ℓ} is a seminorm on $\mathcal{A}$. The topology generated by the collection of seminorms $\{p_{\ell} \mid \ell \in \mathcal{A}^*\}$ is called the *weak topology* on $\mathcal{A}$, and is denoted wk .

Definition 2.8. For $x \in \mathcal{A}$, define $r_x(\ell) = |\ell(x)|$. This is a seminorm on $\mathcal{A}^*$. The topology generated by the collection of seminorms $\{r_x \mid x \in \mathcal{A}\}$ is called the *weak-star topology* on $\mathcal{A}^*$, and is denoted wk^* .

Remark 2.9. Let (X, Ω, μ) be a measure space. Then, the dual space of $L^\infty(\mu)$ is $L^1(\mu)$ with seminorms defined by

$$\rho_f(g) = \left| \int_X fg d\mu \right|$$

for $f \in L^1(\mu)$ and $g \in L^\infty(\mu)$. By Holder's inequality, we have $\rho_f(g) \leq \|fg\|_1 \leq \|f\|_1 \|g\|_\infty$, and so these seminorms are well-defined.

These topologies can be defined on any normed space. In addition, for a Hilbert space $\mathcal{H}$, we can define the following topologies on $\mathcal{B}(\mathcal{H})$:

Definition 2.10. For $f, g \in \mathcal{H}$, define $\rho_{f,g}: \mathcal{B}(\mathcal{H}) \rightarrow \mathbb{R}$ by $\rho_{f,g}(A) = |\langle Af, g \rangle|$. The *weak operator topology* (WOT) is the topology generated by the collection of seminorms

$$\{\rho_{f,g} \mid f, g \in \mathcal{H}\}$$

Later on, as part of the functional calculus, we will use the weak operator topology. Nevertheless though, I believe it's imported to also mention the strong operator topology because of their similar definitions. Even so, these topologies are different. To see this, consider the following two examples on $\ell^2(\mathbb{N})$:

Example 2.11. Let $P_k: \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ be the projection onto the orthogonal complement of the first k components, i.e. for $(a_n)_{n \in \mathbb{N}}$,

$$P_k(a_n)_{n \in \mathbb{N}} = (0, \dots, 0, a_{k+1}, a_{k+2}, \dots)$$

This is a bounded operator as

$$\begin{aligned} \|P_k(a_n)_{n \in \mathbb{N}}\|^2 &= \sum_{n \in \mathbb{N}} |a_{n+k}|^2 \\ &\leq \sum_{n \in \mathbb{N}} |a_n|^2 \\ &= \|(a_n)_{n \in \mathbb{N}}\|^2 \end{aligned}$$

Along with this, $P_k(a_n)_{n \in \mathbb{N}}$ converges to 0 as $k \rightarrow \infty$. Thus, P_k converges strongly to 0. However, for $m < k$ and $(a_n)_{n \in \mathbb{N}} \in \ell^2(\mathbb{N})$, we have

$$\begin{aligned} (P_k - P_m)(a_n)_{n \in \mathbb{N}} &= P_m(a_n)_{n \in \mathbb{N}} - P_k(a_n)_{n \in \mathbb{N}} \\ &= (0, \dots, a_{m+1}, a_{m+2}, \dots) - (0, \dots, a_{k+1}, a_{k+2}, \dots) \\ &= (0, \dots, a_{m+1}, a_{m+2}, \dots, a_{k-1}, a_k, 0, 0, \dots) \end{aligned}$$

thus for all $m < k$, there exists a sequence $(a_n)_{n \in \mathbb{N}}$ such that $(P_m - P_k)(a_n)_{n \in \mathbb{N}} = (a_n)_{n \in \mathbb{N}}$. Therefore, P_k does not converge in norm to 0. This shows that the strong operator topology, despite the name, is weaker than the norm topology.

Example 2.12. Next, consider the shift operator $R : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ defined as

$$R(a_1, a_2, \dots) = (0, a_1, a_2, \dots)$$

For sequences $(a_n)_{n \in \mathbb{N}}, (b_n)_{n \in \mathbb{N}}$, we have

$$\begin{aligned} \left| \left\langle R^k(a_n)_{n \in \mathbb{N}}, (b_n)_{n \in \mathbb{N}} \right\rangle \right| &= \left| \sum_{n \in \mathbb{N}} a_n \bar{b}_{n+k} \right| \\ &= \left| \sum_{n \in \mathbb{N}} a_n \overline{P_{k-1}(b_n)_{n \in \mathbb{N}}} \right| \\ &= \left\| R^k(a_n)_{n \in \mathbb{N}} \right\| \left\| P_{k-1}(b_n)_{n \in \mathbb{N}} \right\| \\ &= \left\| (a_n)_{n \in \mathbb{N}} \right\| \left\| P_{k-1}(b_n)_{n \in \mathbb{N}} \right\| \end{aligned}$$

From the previous example, we have $\|P_{k-1}(b_n)_{n \in \mathbb{N}}\|$ goes to 0 as $k \rightarrow \infty$. Thus R^k goes to 0 under the weak operator topology. But no positive power of R removes any components of the sequences it acts on, and so R^k does not converge to 0 in the strong operator topology.

With these two examples, we have that weak operator convergence is weaker than strong operator convergence, which itself is weaker than convergence in norm. Of course there are many more things we could explore with these operator topologies, such as how limits of products of operators behave in each of these topologies, but that's for another time. For now, we want to continue building up what we need to build a stronger understanding of normal operators.

Definition 2.13. For $f \in \mathcal{H}$, define $\rho_f(A) = \|Af\|_{\mathcal{H}}$. The topology on $\mathcal{B}(\mathcal{H})$ generated by these seminorms is called the *strong operator topology* (SOT).

Remark 2.14. For an operator $N \in \mathcal{B}(H)$, we defined the C^* algebra $C^*(N)$, which uses the operator norm. In a similar fashion, we can define a C^* algebra using the weak operator seminorms. This algebra is called the von Neumann algebra of N , denoted $W^*(N)$, and is the WOT-closure of polynomials in N and N^* .

2.2 Operator-Valued Measures

Let X be any set, and let Ω be a σ -algebra over X . Areal measure is a function $\mu : \Omega \rightarrow [0, \infty]$ that is countably additive and $\mu(\emptyset) = 0$. In our exploration of normal operators, we require measures which output projections on a Hilbert space $\mathcal{H}$.

Definition 2.15. Let X be a space with σ -algebra Ω , and let $\mathcal{H}$ be a Hilbert space. A *spectral* or *projection-valued measure* on the triple $(X, \Omega, \mathcal{H})$ is a map $E : \Omega \rightarrow \mathcal{B}(\mathcal{H})$ such that

- (i) $E(S)$ is a projection for each $S \in \Omega$
- (ii) $E(X) = I_{\mathcal{H}}$ and $E(\emptyset) = 0$
- (iii) $E(S_1 \cap S_2) = E(S_1)E(S_2)$ for $S_1, S_2 \in \Omega$

(iv) E is countably additive, i.e.

$$E\left(\bigcup_{n \in \mathbb{N}} S_n\right) = \sum_{n \in \mathbb{N}} E(S_n)$$

for a pairwise disjoint collection $\{S_n \mid n \in \mathbb{N}\} \subseteq \Omega$

Careful consideration needs to be made here as products of projections may not be projections. This can be resolved by noticing that

$$\begin{aligned} E(S_1)E(S_2) &= E(S_1 \cap S_2) \\ &= E(S_2 \cap S_1) \\ &= E(S_2)E(S_1) \end{aligned}$$

and using that the product of two projections is a projection if and only if they commute. Projection-valued measures are just a small portion of a larger class of operator-valued measures. These measures play a particularly important role in the theory of quantum information [1]

Remark 2.16. For a normal operator $N \in \mathcal{B}(\mathcal{H})$, $\sigma(N)$ is a topological space. The Borel σ -algebra of subsets of $\sigma(N)$, denoted $\text{Borel}(\sigma(N))$, is the smallest σ -algebra on $\sigma(N)$ containing the topology.

Definition 2.17. Let Ω be a σ -algebra on a set X . Let μ, ν be complex valued measures on Ω . We say that ν is absolutely continuous with respect to μ , writing $\nu \ll \mu$ if $\nu(S) = 0$ whenever $\mu(S) = 0$ for any μ -measurable set $S \in \Omega$.

Remark 2.18. Let (X, Ω, μ) be a measure space with complex-valued measure μ , and let $f : X \rightarrow \mathbb{C}$. Then, we can define a measure $\nu = f\mu$ by

$$\nu(S) = \int_S f d\mu$$

Moreover, ν is absolutely continuous with respect to μ . The Radon-Nikodym theorem below gives a converse to this statement.

Theorem 2.19. *Radon-Nikodym Theorem*

Given a measure space (X, Ω) , and σ -finite measures μ, ν on Ω . If ν is absolutely continuous with respect to μ , then there is an integrable function $f : X \rightarrow \mathbb{C}$ such that

$$\nu(S) = \int_S f d\mu$$

for any measurable $A \in \Omega$. f is called the Radon-Nikodym derivative of ν with respect to μ , and denoted $f = \frac{d\nu}{d\mu}$. Moreover, $f \in L^1(\mu)$

We won't prove this theorem here, but will make use of Radon-Nikodym derivatives later.

3 The Spectral Theorem for Bounded Normal Operators

Now that we have a fair bit of background, we can start using the results in the context of normal operators. An important distinction to be ware of is that of the point spectrum and the full spectrum of a normal operator.

Definition 3.1. Let $N \in \mathcal{B}(\mathcal{H})$ be a normal operator. The *point spectrum* $\sigma_p(N)$ is defined as the set of explicit eigenvalues of N :

$$\sigma_p(N) = \{\lambda \in \mathbb{C} \mid \ker(N - \lambda I_{\mathcal{H}}) \neq 0\}$$

As $\ker(N - \lambda I_{\mathcal{H}}) \neq 0$, $N - \lambda I_{\mathcal{H}}$ is not invertible.

Definition 3.2. The *full spectrum*, or simply just the *spectrum* $\sigma(N)$ is

$$\{z \in \mathbb{C} \mid N - zI_{\mathcal{H}} \text{ is not boundedly invertible}\}$$

$N - zI_{\mathcal{H}}$ not being boundedly invertible, one of two possibilities are present. Either

- i) $N - zI_{\mathcal{H}}$ is not invertible, or
- ii) $(N - zI_{\mathcal{H}})^{-1}$ exists, but is unbounded

The first condition implies $\sigma_p(N) \subseteq \sigma(N)$. The second condition implies that values in $\sigma(N)$ are not necessarily eigenvalues of N . We will see this more explicitly with examples later on.

Remark 3.3. For finite dimensional spaces $\mathcal{H}$, all linear operators are bounded. As a result, for any normal operator $N : \mathcal{H} \rightarrow \mathcal{H}$ with $\mathcal{H}$ finite dimensional, the point spectrum and full spectrum coincide, $\sigma_p(N) = \sigma(N)$.

To find examples of normal operators for which the point spectrum and full spectrum are distinct, we will have to look at operators on infinite dimensional spaces.

Remark 3.4. Along with the point and full spectra of an operator, there are many more types of operator spectra. These include the approximate, continuous, and residual spectra [4]. While these spectra won't be of significant importance here, mentioning them here is to emphasize that the spectrum of an operator extends beyond just the eigenvalues.

Example 3.5. Consider the operator $A : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ defined by

$$A(a_n)_{n \in \mathbb{N}} = \left(\frac{a_n}{n} \right)_{n \in \mathbb{N}}$$

The eigenvalues of A are $1/n$ for $n \in \mathbb{N}$. As well, A is injective, and so 0 is not an eigenvalue of A . However, I claim that A is not surjective. To see this, notice that A is injective. Consider the sequence $(\frac{1}{n})_{n \in \mathbb{N}}$. We have

$$\begin{aligned} \left\| \left(\frac{1}{n} \right)_{n \in \mathbb{N}} \right\|^2 &= \left\langle \left(\frac{1}{n} \right)_{n \in \mathbb{N}}, \left(\frac{1}{n} \right)_{n \in \mathbb{N}} \right\rangle \\ &= \sum_{n \in \mathbb{N}} \left(\frac{1}{n} \right) \overline{\left(\frac{1}{n} \right)} \\ &= \sum_{n \in \mathbb{N}} \frac{1}{n^2} = \frac{\pi^2}{6} \end{aligned}$$

and so $(\frac{1}{n})_{n \in \mathbb{N}} \in \ell^2(\mathbb{N})$. If A were surjective, then there would be a sequence $(a_n)_{n \in \mathbb{N}} \in \ell^2(\mathbb{N})$ such that $A(a_n)_{n \in \mathbb{N}} = (\frac{1}{n})_{n \in \mathbb{N}}$. A is injective on $\ell^2(\mathbb{N})$. It is left to the reader to verify that the extension of A to all $\mathbb{N}$ -index sequences remains injective. With this, there is only one sequence that could satisfy this, and this sequence is that which has 1

in every term, $(1)_{n \in \mathbb{N}}$. However, this sequence is not in $\ell^2(N)$. As a result, $(\frac{1}{n})_{n \in \mathbb{N}}$ is an $\ell^2(\mathbb{N})$ sequence not contained in the image of A , i.e. A is not surjective. Because of this, $0 \in \sigma(A)$ even though 0 is not an eigenvalue of A .

In fact, we can have normal operators N for which $\sigma_p(N) = \emptyset$ and $\sigma(N) \neq \emptyset$. We will see an example on this later.

Theorem 3.6. *The Spectral Theorem for Normal Operators. Let $N \in \mathcal{B}(\mathcal{H})$ be a normal operator. There is a unique projection-valued measure E on $\text{Borel}(\sigma(N))$ such that*

1. $N = \int_{\sigma(N)} I_{\sigma(N)} dE$
2. $E(S) \neq 0$ for nonempty open subsets $S \subseteq \sigma(N)$
3. $AN = NA$ and $AN^* = N^*A$ if and only if $AE(S) = E(S)A$ for any $S \subseteq \sigma(N)$ and any $A \in \mathcal{B}(\mathcal{H})$

Here, $I_{\sigma(N)} : \sigma(N) \rightarrow \mathbb{C}$ is the inclusion map.

This is but one form of the spectral theorem. This form is analogous to the decomposition of a finite dimensional normal operator N into a weighted sum of mutually orthogonal projections:

$$N = \sum_{\lambda \in \sigma(N)} \lambda P_\lambda$$

where P_λ is the projection onto the λ -eigenspace. Additionally, we have $P_\lambda P_\phi = 0$ for distinct $\lambda, \phi \in \sigma(N)$.

In order to prove the spectral theorem for normal operators, we first prove the following theorem [11]

Theorem 3.7. *Let $N \in \mathcal{B}(\mathcal{H})$ be a normal operator, and $C^*(N)$ the C^* -algebra generated by N . The set*

$$\Delta = \{f : C^*(N) \rightarrow \mathbb{C} \mid f \neq 0 \text{ is a homomorphism}\}$$

is called the maximal ideal space of $C^(N)$.*

Recall that $C^(N)$ is the norm closure of polynomials in N and N^* . Moreover, $C^*(N)$ is a commutative subalgebra of $\mathcal{B}(\mathcal{H})$. As a result [11], we can identify Δ with $\sigma(N)$ so that $G(N) = I_{\sigma(N)}$ where $G : C^*(N) \rightarrow C(\sigma(N))$ is the Gelfand transform.*

Then there is a unique projection-valued measure $E : \text{Borel}(\sigma(N)) \rightarrow \mathcal{B}(\mathcal{H})$ such that

$$T = \int_{\sigma(N)} G(T) dE$$

for any $T \in C^(N)$.*

Through the identification $\Delta = \sigma(N)$, we are able to show existence of the projection-valued measure using theorem 3.7. With the existence confirmed, we are then able to prove the rest of the spectral theorem.

Proof. of Theorem 3.7 As $C^*(N)$ is a commutative C^* -algebra, the Gelfand-Naimark theorem states that G is an isometric $*$ -isomorphism. For $f, g \in \mathcal{H}$, let $\Psi_{f,g} : C(\sigma(N)) \rightarrow \mathbb{C}$ be the map defined by

$$\Psi_{f,g}(\phi) = \langle G^{-1}(\phi)f, g \rangle$$

$\Psi_{f,g}$ is a linear functional. Then by the Riesz-Markov theorem [6] there is a unique complex-valued regular Borel measure $E_{f,g}$ defined on subsets of $\sigma(N)$ such that

$$\Psi_{f,g}(\phi) = \int_{\sigma(N)} \phi dE_{f,g}$$

Now that we have a measure, we can extend $\Psi_{f,g}$ to bounded Borel measurable functions on $\sigma(N)$, for which the integral with respect to $E_{f,g}$ is well-defined. Moreover, the map

$$\tilde{\Psi}_{f,g}(\gamma) = \int_{\sigma(N)} \gamma dE_{f,g}$$

bounded as

$$\begin{aligned} |\Psi_{f,g}(\gamma)| &= \left| \int_{\sigma(N)} \gamma dE_{f,g} \right| \\ &\leq \int_{\sigma(N)} |\gamma| d|E_{f,g}| \\ &\leq \|\gamma\|_{\infty} \int_{\sigma(N)} d|E_{f,g}| \\ &= \|\gamma\|_{\infty} |\langle f, g \rangle| \\ &= \|\gamma\|_{\infty} \|f\|_{\mathcal{H}} \|g\|_{\mathcal{H}} \end{aligned}$$

Because of this, the map $(f, g) \mapsto E_{f,g}$ is bounded and sesquilinear. and so there is a map Θ defined for bounded functions $\gamma : \sigma(N) \rightarrow \mathbb{C}$ so that

$$\langle \Theta(\gamma)f, g \rangle = \int_{\sigma(N)} \gamma dE_{f,g}$$

From this, we see that Θ is an extension of G^{-1} from $C(\sigma(N))$ to bounded functions on $\sigma(N)$. Then, define $E : \text{Borel}(\sigma(N)) \rightarrow \mathcal{B}(\mathcal{H})$ by $E(S) = \Theta(\chi_S)$. I claim that E is a projection-valued measure. To see this, notice the following

$$\begin{aligned} \langle E(\sigma(N))f, g \rangle &= \langle \Theta(\chi_{\sigma(N)})f, g \rangle \\ &= \int_{\sigma(N)} \chi_{\sigma(N)} dE_{f,g} \\ &= \int_{\sigma(N)} dE_{f,g} \\ &= \langle f, g \rangle \end{aligned}$$

$$\begin{aligned}
\langle E(\emptyset)f, g \rangle &= \langle \Theta(\chi_\emptyset)f, g \rangle \\
&= \int_{\sigma(N)} \chi_\emptyset dE_{f,g} \\
&= \int_{\sigma(N)} 0 dE_{f,g} \\
&= 0
\end{aligned}$$

For $S_1, S_2 \in \text{Borel}(\sigma(N))$

$$\begin{aligned}
\langle E(S_1 \cap S_2)f, g \rangle &= \langle \Theta(\chi_{S_1 \cap S_2})f, g \rangle \\
&= \langle \Theta(\chi_{S_1} \chi_{S_2})f, g \rangle \\
&= \langle \Theta(\chi_{S_1})\Theta(\chi_{S_2})f, g \rangle \\
&= \langle E(S_1)E(S_2)f, g \rangle
\end{aligned}$$

and for a pairwise disjoint collection $\{S_n \mid n \in \mathbb{N}\} \subseteq \text{Borel}(\sigma(N))$

$$\begin{aligned}
\left\langle E\left(\bigcup_{n \in \mathbb{N}} S_n\right)f, g \right\rangle &= \left\langle \Theta\left(\chi_{\bigcup_{n \in \mathbb{N}} S_n}\right)f, g \right\rangle \\
&= \left\langle \Theta\left(\sum_{n \in \mathbb{N}} \chi_{S_n}\right)f, g \right\rangle \\
&= \left\langle \sum_{n \in \mathbb{N}} \Theta(\chi_{S_n})f, g \right\rangle \\
&= \left\langle \sum_{n \in \mathbb{N}} E(S_n)f, g \right\rangle
\end{aligned}$$

Lastly, for any $S \in \text{Borel}(\sigma(N))$, and any $f, g \in \mathcal{H}$ we have

$$\begin{aligned}
\langle E(S)f, g \rangle &= \langle \Theta(\chi_S)f, g \rangle \\
&= \int_{\sigma(N)} \chi_S dE_{f,g} \\
&= \tilde{\Psi}_{f,g}(\chi_S)
\end{aligned}$$

Uniqueness of E follows from uniqueness of $E_{f,g}$ for any $f, g \in \mathcal{H}$.

□

Now to finish the proof of the spectral theorem:

Proof. Through the association we made at the start, that $G(N) = I_{\sigma(N)}$, we have

$$N = \int_{\sigma(N)} I_{\sigma(N)} dE$$

Thus the first statement of the spectral theorem is proven. To prove the third statement, let $T \in C^*(N)$ and $U \in \mathcal{B}(\mathcal{H})$ and $S \in \text{Borel}(\sigma(N))$, then

$$\begin{aligned}\langle UTf, g \rangle &= \langle Tf, U^*g \rangle \\ &= \int_{\sigma(N)} G(T) dE_{f, U^*g}\end{aligned}$$

and

$$\langle TUF, g \rangle = \int_{\sigma(N)} G(T) dE_{Uf, g}$$

If T and U commute for any $T \in C^*(N)$, then the measures E_{f, U^*g} and $E_{Uf, g}$ are the same, and so $E(S)U = UE(S)$ for any Borel set $S \subseteq \sigma(N)$. Moreover, the condition that U commutes with every element of $C^*(N)$ can be reduced down to commuting with N and N^* as $C^*(N)$ is generated by N and N^* . Conversely,

$$\begin{aligned}\langle UE(S)f, g \rangle &= \langle E(S)f, U^*g \rangle \\ &= E_{f, U^*g}(S)\end{aligned}$$

and

$$\langle E(S)Uf, g \rangle = E_{Uf, g}(S)$$

and so if $E(S)$ and U commute for all $S \in \text{Borel}(\sigma(N))$, the measures $E_{Uf, g}$ and E_{f, U^*g} are the same, and so U commutes with N and N^* .

For the second statement of the spectral theorem, let $S \subseteq \sigma(N)$ be a nonempty open set. Then, the characteristic function χ_S is nonzero. As $C(\sigma(N))$ is contained in the collection of bounded functions on $\sigma(N)$, we can find a nonzero continuous function $\phi : \sigma(N) \rightarrow \mathbb{C}$ whose support is contained in S . By applying the appropriate scaling, we can have $|\phi(t)| \leq \chi_S(t) = 1$ for all $t \in S$. As the Gelfand transform is an isometry between $C^*(N)$ and $C(\sigma(N))$, $G^{-1}(\phi)$ is a nonzero operator on $\mathcal{B}(\mathcal{H})$, and so there exists $f, g \in \mathcal{H}$ such that

$$\begin{aligned}|\langle E(S)f, g \rangle| &= |\langle \Theta(\chi_S)f, g \rangle| \\ &= |\langle \Theta(\phi)f, g \rangle| \\ &\leq |\langle G^{-1}(\phi)f, g \rangle| > 0\end{aligned}$$

Therefore, $E(S) \neq 0$. Notice that this holds as Θ is an extension of G^{-1} . This concludes the proof of the spectral theorem. □

We would like the PVM E to behave similarly. In particular, for an arbitrary Hilbert space $\mathcal{H}$ and normal $N \in \mathcal{B}(\mathcal{H})$, we would like $E(\{\lambda\}) \neq 0$ for $\lambda \in \sigma_p(N)$.

If $\lambda \in \sigma_p(N)$, there is a nonzero $f \in \mathcal{H}$ such that $Nf = \lambda f$. Now, by the spectral theorem

$$\begin{aligned} \lambda f &= Nf \\ &= \left(\int_{\sigma(N)} I_{\sigma(N)} dE \right) f \\ &= \left(\int_{\sigma(N) \setminus \{\lambda\}} I_{\sigma(N)} dE + \int_{\{\lambda\}} I_{\sigma(N)} dE \right) f \end{aligned}$$

Focusing on the second integral, we have

$$\begin{aligned} \int_{\{\lambda\}} I_{\sigma(N)} dE &= \int_{\sigma(N)} I_{\sigma(N)} \chi_{\{\lambda\}} dE \\ &= \lambda \int_{\sigma(N)} \chi_{\{\lambda\}} dE \\ &= \lambda E(\{\lambda\}) \end{aligned}$$

In the integral over $\sigma(N) \setminus \{\lambda\}$, the map $I_{\sigma(N)}$ never takes on the value λ , and so this integral must send f to 0. If this were not the case, f would not be a λ -eigenvector of N . Then, $E\{\lambda\}f = f \neq 0$, i.e. f is a nonzero element in the range of $E(\{\lambda\})$. Thus, $E(\{\lambda\}) \neq 0$.

Proposition 3.8. *In fact, the map $\phi \mapsto \int_{\sigma(N)} \phi dE$ is a homomorphism.*

Proof. For this, let $\phi, \gamma : \sigma(N) \rightarrow \mathbb{C}$ be simple functions. Then, we can write $\phi = \sum_{k=1}^n a_k \chi_{S_k}$ and $\gamma = \sum_{\ell=1}^m b_\ell \chi_{Q_\ell}$ for complex numbers a_k, b_ℓ and measurable sets S_k and Q_ℓ . Integrating the product $\phi\gamma$:

$$\begin{aligned}
\int_{\sigma(N)} \phi \gamma dE &= \int_{\sigma(N)} \left(\sum_{k=1}^n a_k \chi_{S_k} \right) \left(\sum_{\ell=1}^m b_\ell \chi_{Q_\ell} \right) dE \\
&= \int_{\sigma(N)} \sum_{k=1}^n \sum_{\ell=1}^m a_k b_\ell \chi_{S_k} \chi_{Q_\ell} dE \\
&= \sum_{k=1}^n \sum_{\ell=1}^m a_k b_\ell \int_{\sigma(N)} \chi_{S_k \cap Q_\ell} dE \\
&= \sum_{k=1}^n \sum_{\ell=1}^m a_k b_\ell E(S_k \cap Q_\ell) \\
&= \sum_{k=1}^n \sum_{\ell=1}^m a_k b_\ell E(S_k) E(Q_\ell) \\
&= \left(\sum_{k=1}^n a_k E(S_k) \right) \left(\sum_{\ell=1}^m b_\ell E(Q_\ell) \right) \\
&= \left(\sum_{k=1}^n \int_{\sigma(N)} a_k \chi_{S_k} dE \right) \left(\sum_{\ell=1}^m \int_{\sigma(N)} b_\ell \chi_{Q_\ell} dE \right) \\
&= \left(\int_{\sigma(N)} \sum_{k=1}^n a_k \chi_{S_k} dE \right) \left(\int_{\sigma(N)} \sum_{\ell=1}^m b_\ell \chi_{Q_\ell} dE \right) \\
&= \left(\int_{\sigma(N)} \phi dE \right) \left(\int_{\sigma(N)} \gamma dE \right)
\end{aligned}$$

By taking limits of simple functions, we obtain that for any bounded functions $\phi, \gamma : \sigma(N) \rightarrow \mathbb{C}$, we have

$$\int_{\sigma(N)} \phi \gamma dE = \left(\int_{\sigma(N)} \phi dE \right) \left(\int_{\sigma(N)} \gamma dE \right)$$

□

Now that we have the spectral theorem, let's spend a bit of time looking at a few examples. The purpose of this is to give concrete examples of how, for a normal operator $N \in \mathcal{B}(\mathcal{H})$, the spectrum $\sigma(N)$ and point $\sigma_p(N)$ differ.

Example 3.9. To begin, we look at the following matrix. Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be the linear

operator defined by the matrix

$$\begin{bmatrix} 43 & -\frac{34\sqrt{5}}{25} & \frac{62\sqrt{5}}{25} & 8 \\ -\frac{34\sqrt{5}}{25} & \frac{896}{25} & \frac{272}{25} & -\frac{44\sqrt{5}}{25} \\ \frac{62\sqrt{5}}{25} & \frac{272}{25} & \frac{1154}{25} & -\frac{8\sqrt{5}}{25} \\ 8 & -\frac{44\sqrt{5}}{25} & -\frac{8\sqrt{5}}{25} & 46 \end{bmatrix}$$

It is left to the reader to verify that

- (i) T is normal
- (ii) The spectrum of T is $\sigma(T) = \{27, 36, 54\}$
- (iii) The eigenspaces are

$$\begin{aligned} \mathcal{E}_{27} &= \text{span} \left\{ \vec{v}_{27} = \begin{bmatrix} \frac{1}{3} & \frac{26\sqrt{5}}{75} & -\frac{6\sqrt{5}}{25} & 0 \end{bmatrix}^T \right\} \\ \mathcal{E}_{36} &= \text{span} \left\{ \vec{v}_{36} = \begin{bmatrix} \frac{2}{3} & -\frac{11\sqrt{5}}{75} & -\frac{2\sqrt{5}}{75} & -\frac{2}{3} \end{bmatrix}^T \right\} \\ \mathcal{E}_{54} &= \text{span} \left\{ \vec{v}_{54,1} = \begin{bmatrix} 0 & \frac{6\sqrt{5}}{25} & \frac{26\sqrt{5}}{75} & -\frac{1}{3} \end{bmatrix}^T, \vec{v}_{54,2} = \begin{bmatrix} -\frac{2}{3} & \frac{2\sqrt{5}}{75} & -\frac{11\sqrt{5}}{75} & -\frac{2}{3} \end{bmatrix}^T \right\} \end{aligned}$$

- (iv) The set of vectors

$$\left\{ \begin{bmatrix} \frac{1}{3} \\ \frac{26\sqrt{5}}{75} \\ -\frac{6\sqrt{5}}{25} \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{2}{3} \\ -\frac{11\sqrt{5}}{75} \\ -\frac{2\sqrt{5}}{75} \\ -\frac{2}{3} \end{bmatrix}, \begin{bmatrix} 0 \\ \frac{6\sqrt{5}}{25} \\ \frac{26\sqrt{5}}{75} \\ -\frac{1}{3} \end{bmatrix}, \begin{bmatrix} -\frac{2}{3} \\ \frac{2\sqrt{5}}{75} \\ -\frac{11\sqrt{5}}{75} \\ -\frac{2}{3} \end{bmatrix} \right\}$$

is orthonormal

All linear operators on finite dimensional spaces are bounded. As a result, the full and point spectra coincide. Define the projection-valued measure $E : \mathcal{P}(\sigma(T)) \rightarrow \mathcal{B}(\mathcal{H})$ so that

$$\begin{aligned} E(\emptyset)\vec{u} &= 0 \\ E(\{27\})\vec{u} &= P_{27}\vec{u} = (\vec{v}_{27} \cdot \vec{u})\vec{v}_{27} \\ E(\{36\})\vec{u} &= P_{36}\vec{u} = (\vec{v}_{36} \cdot \vec{u})\vec{v}_{36} \\ E(\{54\})\vec{u} &= P_{54}\vec{u} = (\vec{v}_{54,1} \cdot \vec{u})\vec{v}_{54,1} + (\vec{v}_{54,2} \cdot \vec{u})\vec{v}_{54,2} \end{aligned}$$

for all $\vec{u} \in \mathbb{R}^4$, and extending so that $E(S_1 \cap S_2) = E(S_1)E(S_2)$ for any subsets $S_1, S_2 \subseteq \sigma(T)$ and $E(S_1 \cup S_2) = E(S_1) + E(S_2)$ for disjoint subsets $S_1, S_2 \subseteq \sigma(T)$. Notice that this makes E a discrete measure. Then, taking $I_{\sigma(T)} : \sigma(T) \rightarrow \mathbb{R}$ to be the inclusion map, we

have

$$\begin{aligned}
 \int_{\sigma(T)} I_{\sigma(T)} dE &= \int_{\sigma(T)} (27\chi_{\{27\}} + 36\chi_{\{36\}} + 54\chi_{\{54\}}) dE \\
 &= 27 \int_{\sigma(T)} \chi_{\{27\}} dE + 36 \int_{\sigma(T)} \chi_{\{36\}} dE + 54 \int_{\sigma(T)} \chi_{\{54\}} dE \\
 &= 27E(\{27\}) + 36E(\{36\}) + 54E(\{54\}) \\
 &= 27P_{27} + 36P_{36} + 54P_{54}
 \end{aligned}$$

From this last line, we see that this aligns with the spectral theorem for finite dimensional operators we saw earlier. Here, χ_S is the characteristic function for $S \subseteq \sigma(T)$. This was but one finite dimensional example, but a similar procedure can be used for any finite dimensional normal operator. Thus we conclude that this integral form of the spectral theorem agrees with the finite dimensional case. Now, let's look at a couple examples on infinite dimensional spaces.

Example 3.10. Let $A : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$ be the linear operator defined by as

$$A(a_n)_{n \in \mathbb{N}} = \left(\frac{a_n}{n} \right)_{n \in \mathbb{N}}$$

For any sequences $(a_n)_{n \in \mathbb{N}}, (b_n)_{n \in \mathbb{N}}$, we have

$$\begin{aligned}
 \langle (a_n)_{n \in \mathbb{N}}, A^*(b_n)_{n \in \mathbb{N}} \rangle &= \langle A(a_n)_{n \in \mathbb{N}}, (b_n)_{n \in \mathbb{N}} \rangle \\
 &= \left\langle \left(\frac{a_n}{n} \right)_{n \in \mathbb{N}}, (b_n)_{n \in \mathbb{N}} \right\rangle \\
 &= \sum_{n \in \mathbb{N}} \frac{a_n}{n} \bar{b}_n \\
 &= \sum_{n \in \mathbb{N}} a_n \frac{\bar{b}_n}{n} \\
 &= \left\langle (a_n)_{n \in \mathbb{N}}, \left(\frac{b_n}{n} \right)_{n \in \mathbb{N}} \right\rangle \quad (\text{as } n \in \mathbb{R}) \\
 &= \langle (a_n)_{n \in \mathbb{N}}, A(b_n)_{n \in \mathbb{N}} \rangle
 \end{aligned}$$

Thus, A is self-adjoint, and hence normal. As discussed earlier, the point spectrum of A is

$$\sigma_p(A) = \{1/k \mid k \in \mathbb{N}\}$$

and $0 \in \sigma(A)$ but $0 \notin \sigma_p(A)$. Next, we want to define the projection-valued measure E for this operator. Starting with the point spectrum, for each $n \in \mathbb{N}$, define E on singletons of $\sigma_p(A)$ so that $E(\{1/k\})$ is the projection onto the k th component, and extend to all of $\sigma_p(A)$ to get the desired properties. But wait, $\sigma(A) \neq \sigma_p(A)$. In particular, we still need to define $E(\{0\})$. To do this, notice that for any $k \in \mathbb{N}$

$$\begin{aligned}
 0 &= E(\emptyset) \\
 &= E(\{0\} \cap \{1/k\}) \\
 &= E(\{0\})E(\{1/k\})
 \end{aligned}$$

but $E(\{1/k\})$ is the projection onto the k th component. As k was arbitrary, the above equation must hold for any $k \in \mathbb{N}$. Due to this, we must have $E(\{0\}) = 0$. As in the finite dimensional case, this projection measure E is also discrete, the difference here being that we get an infinite sum:

$$\begin{aligned} \int_{\sigma(A)} I_{\sigma(A)} dE &= \int_{\sigma(A)} \left(0\chi_{\{0\}} + \sum_{n \in \mathbb{N}} \frac{1}{n} \chi_{\{1/n\}} \right) dE \\ &= \sum_{n \in \mathbb{N}} \left(\frac{1}{n} \int_{\sigma(A)} \chi_{\{1/n\}} dE \right) \\ &= \sum_{n \in \mathbb{N}} \frac{1}{n} E(\{1/n\}) \\ &= \sum_{n \in \mathbb{N}} \frac{1}{n} P_n \\ &= A \end{aligned}$$

where P_n is the projection onto the n th component.

With these two examples, we've been able to write the integral as a sum due to the discrete measures. However, this is not always the case, as we will see with the shift operator later. In order to understand the shift operator, we first need to look at two other linear operators, which will come in handy when dealing with the shift operator.

Example 3.11. Let $\mathcal{F} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ be the Fourier transform;

$$(\mathcal{F}f)(\omega) = \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t} dt$$

By Plancherel's theorem, the Fourier transform is a unitary operator. The Fourier transform also has a shift property, namely for $a \in \mathbb{R}$

$$\begin{aligned} \mathcal{F}(f(t-a)) &= \int_{\mathbb{R}} f(t-a) e^{-2\pi i \omega t} dt \\ &= \int_{\mathbb{R}} f(t) e^{-2\pi i \omega (t+a)} dt \\ &= e^{-2\pi i \omega a} \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t} dt \\ &= e^{-2\pi i \omega a} \mathcal{F}(f(t)) \end{aligned}$$

Along with this, a calculation shows that for $f(t) = e^{-\pi t^2}$,

$$(\mathcal{F}f)(\omega) = e^{-\pi \omega^2}$$

Thus, 1 is an eigenvalue of $\mathcal{F}$. Another important property of the Fourier transform is the relation it has to derivatives: For differentiable $f : \mathbb{R} \rightarrow \mathbb{C}$

$$\begin{aligned} (\mathcal{F}f') &= \int_{\mathbb{R}} f'(t) e^{-2\pi i \omega t} dt \\ &= \lim_{L \rightarrow \infty} (f(t) e^{-2\pi i \omega t}) \Big|_{t=-L}^L + 2\pi i \omega \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t} dt \end{aligned}$$

If $f \in L^2(\mathbb{R})$, the limit goes to 0 leaving only the integral on the right. Applying this to the function $h(t) = te^{-\pi t^2}$:

$$\begin{aligned}
 (\mathcal{F}h)(\omega) &= \int_{\mathbb{R}} te^{-\pi t^2} e^{-2\pi i\omega t} dt \\
 &= \int_{\mathbb{R}} \left(\frac{-1}{2\pi} \frac{d}{dt} e^{-\pi t^2} \right) e^{-2\pi i\omega t} dt \\
 &= \frac{-2\pi i\omega}{2\pi} \int_{\mathbb{R}} e^{-\pi t^2} e^{-2\pi i\omega t} dt \\
 &= -i\omega \int_{\mathbb{R}} e^{-\pi t^2} e^{-2\pi i\omega t} dt \\
 &= -i\omega e^{-\pi\omega^2} \\
 &= -ih(\omega)
 \end{aligned}$$

Repeating this process, we get that the point spectrum of $\mathcal{F}$ is $\sigma_p(\mathcal{F}) = \{1, -i, -1, i\}$, and the corresponding normalized eigenvectors are of the form

$$h_{k+4n}(t) = \left(\frac{1}{2\pi} \right)^{k+4n} \frac{d^{k+4n}}{dt^{k+4n}} e^{-\pi t^2}$$

where $k = 0, 1, 2, 3$ and $n \in \mathbb{N}_0$. The astute might notice that these are the Hermite polynomials multiplied by a Gaussian factor. [10] Moreover, these eigenvectors form a complete orthonormal basis of $L^2(\mathbb{R})$. The corresponding projections are defined as

$$P_k f = \sum_{n=0}^{\infty} h_{k+4n} \langle f, h_{k+4n} \rangle$$

for $k = 0, 1, 2, 3$. From this, we can define the projection valued measure J by $J(\{i^k\}) = P_k$

Example 3.12. Let $g : \mathbb{R} \rightarrow \mathbb{C}$ be a bounded function. Note that g may not be in $L^2(\mathbb{R})$. Nevertheless, we can define the operator $M_g : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ by

$$(M_g f)(t) = f(t)g(t)$$

To verify that this does indeed work, notice that for $f \in \mathcal{H}$:

$$\begin{aligned}
 \|M_g f\|^2 &= \int_{\mathbb{R}} |gf|^2 dt \\
 &\leq \int_{\mathbb{R}} |\sup(g)|^2 |f|^2 dt \\
 &= |\sup(g)|^2 \int_{\mathbb{R}} |f|^2 dt \\
 &= |\sup(g)|^2 \|f\|^2
 \end{aligned}$$

Thus M_g is a bounded operator as $|\sup(g)| < \infty$. Let $f, h \in L^2(\mathbb{R})$, then

$$\begin{aligned}\langle f, M_g^* h \rangle &= \langle M_g f, h \rangle \\ &= \int_{\mathbb{R}} g(t) f(t) \overline{h(t)} dt \\ &= \int_{\mathbb{R}} f(t) \overline{g(t)} \overline{h(t)} dt \\ &= \langle f, M_{\bar{g}} h \rangle\end{aligned}$$

Thus, $M_g^* = M_{\bar{g}}$. As pointwise multiplication is commutative, we have that M_g is normal. Now let's find the spectrum of M_g . For $f \in L^2(\mathbb{R})$, we see

$$\begin{aligned}(M_g - \lambda I_{L^2(\mathbb{R})})f &= gf - \lambda f \\ &= (g - \lambda)f \\ &= M_{g-\lambda}f\end{aligned}$$

As f was arbitrary, $M_g - \lambda I_{L^2(\mathbb{R})} = M_{g-\lambda}$. Provided it exists, $M_{g-\lambda}^{-1} = M_{1/(g-\lambda)}$. With this, we now check when $M_{1/(g-\lambda)}$ is unbounded. For $M_{1/(g-\lambda)}$ to be bounded, there exists a positive $a \in \mathbb{R}$ such that

$$\left| \frac{1}{g(t) - \lambda} \right| < a \text{ a.e.}$$

or equivalently $|g(t) - \lambda| > 1/a$ a.e.. Write $\varepsilon = 1/a$, then $M_{1/(g-\lambda)}$ is bounded whenever $|g(t) - \lambda| > \varepsilon$ a.e. If no such ε exists, $M_{1/(g-\lambda)}$ is unbounded, such as the case when $g(t) = \lambda$. Thus, $\sigma(M_g) = \{\lambda \in \mathbb{C} \mid \nexists \varepsilon \text{ so that } |g(t) - \lambda| > \varepsilon \text{ a.e.}\}$ [5]. Equivalently, this is the set

$$\{\lambda \in \mathbb{C} \mid m(\{t \in \mathbb{R} \mid |g(t) - \lambda| < \varepsilon\}) > 0 \text{ for all } \varepsilon > 0\}$$

This is exactly the essential image of g . This gives us the full spectrum of M_g , but it remains to show what the eigenvalues are. If λ is an eigenvalue of M_g , then there is a nonzero $f \in \mathcal{H}$ such that

$$M_g f = \lambda f$$

More explicitly, f is a λ -eigenvector of M_g if $g(t)f(t) = \lambda f(t)$ a.e. Since $f \neq 0$, there is a set S with positive (possibly infinite) Lebesgue measure such that $f(t) \neq 0$ for all $t \in S$. In general, we need $(g(t) - \lambda)f(t) = 0$. On this set S in particular, we have $g(t) - \lambda = 0$ a.e. And so, $\sigma_p(M_g) \subseteq \{\lambda \in \mathbb{C} \mid m(\{t \in \mathbb{R} \mid g(t) = \lambda\}) > 0\}$. For the other direction, let $\lambda \in \mathbb{C}$ such that $S = \{t \in \mathbb{R} \mid g(t) = \lambda\}$ has positive finite measure. Consider the characteristic function χ_S on S . Notice

$$\begin{aligned}\int_{\mathbb{R}} |\chi_S|^2 dt &= \int_{\mathbb{R}} \chi_S dt \\ &= \int_S dt \\ &= m(S) < \infty\end{aligned}$$

thus, $\chi_S \in L^2(\mathbb{R})$. Moreover,

$$M_g \chi_S = g(t) \chi_S(t) = \begin{cases} g(t) & t \in S \\ 0 & t \notin S \end{cases}$$

but $g(t) = \lambda$ for all $t \in S$, and so $M_g \chi_S = \lambda \chi_S$. This handles the case where S has finite measure, next, consider the case when $m(S) = \infty$. Here, take the function $f(t) = \chi_S(t) e^{-\pi t^2}$, then

$$M_g f = g(t) \chi_S(t) e^{-\pi t^2} = \begin{cases} g(t) e^{-\pi t^2} & t \in S \\ 0 & t \notin S \end{cases}$$

As before, $g(t) = \lambda$ for all $t \in S$ and 0 otherwise, and so $M_g f = \lambda f$. Lastly,

$$\begin{aligned} \|f\|^2 &= \int_{\mathbb{R}} |f(t)|^2 dt \\ &\leq \int_{\mathbb{R}} |f(t)|^2 dt \\ &\leq \int_{\mathbb{R}} e^{-2\pi t^2} dt \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

thus, $f \in L^2(\mathbb{R})$. This covers the case S has infinite measure. Therefore,

$$\sigma_p(M_g) = \{\lambda \in \mathbb{C} \mid m(\{t \in \mathbb{R} \mid g(t) = \lambda\}) > 0\}$$

Onto the spectral theorem, for a subset $S \subseteq \sigma(M_g) = \text{ess im}(g)$, define the projection-valued measure E by

$$E(S)f = M_{\chi_{g^{-1}(S)}}f$$

where $\chi_{g^{-1}(S)}$ is the characteristic function on the preimage of S under g . This might seem a bit odd, but notice that $\chi_{g^{-1}(\text{ess im}(g))} = 1$ a.e., and so for any $f \in L^2(\mathbb{R})$, we have

$$\begin{aligned} E(\sigma(M_g))f &= M_{\chi_{g^{-1}(\text{ess im}(g))}}f \\ &= M_{\chi_{\mathbb{R}}}f \\ &= f \end{aligned}$$

Therefore, $E(\sigma(M_g)) = I_{L^2(\mathbb{R})}$. As well

$$\begin{aligned} E(\emptyset)f &= M_{\chi_{g^{-1}(\emptyset)}}f \\ &= M_{\chi_{\emptyset}}f \\ &= 0f \\ &= 0 \end{aligned}$$

and so $E(\emptyset) = 0$ as f was arbitrary. Let $S_1, S_2 \subseteq \sigma(M_g)$, then

$$\begin{aligned} \|E(S_1 \cap S_2)f\|^2 &= \int_{\mathbb{R}} |\chi_{g^{-1}(S_1 \cap S_2)}(t)f(t)|^2 dt \\ &= \int_{\mathbb{R}} |\chi_{g^{-1}(S_1) \cap g^{-1}(S_2)}(t)f(t)|^2 dt \\ &= \int_{\mathbb{R}} |\chi_{g^{-1}(S_1)}(t)\chi_{g^{-1}(S_2)}(t)f(t)|^2 dt \\ &= \|E(S_1)E(S_2)f\|^2 \end{aligned}$$

Lastly, if $\{S_n\}_{n \in \mathbb{N}}$ is a collection of disjoint subsets of $\sigma(M_g)$, then

$$\begin{aligned} \left\| E\left(\bigcup_{n \in \mathbb{N}} S_n\right)f \right\|^2 &= \int_{\mathbb{R}} |\chi_{g^{-1}(\bigcup_{n \in \mathbb{N}} S_n)}(t)f(t)|^2 dt \\ &= \int_{\mathbb{R}} |\chi_{\bigcup_{n \in \mathbb{N}} g^{-1}(S_n)}(t)f(t)|^2 dt \\ &= \int_{\mathbb{R}} \left| \sum_{n \in \mathbb{N}} \chi_{g^{-1}(S_n)}(t)f(t) \right|^2 dt \\ &= \left\| \sum_{n \in \mathbb{N}} E(S_n)f \right\|^2 \end{aligned}$$

Therefore, E is indeed a projection-valued measure. Finally, we integrate the inclusion $I_{\sigma(M_g)}$ with respect to E :

$$\int_{\sigma(M_g)} I_{\sigma(M_g)} dE$$

If g is a simple function, then $\text{ess im}(g)$ is a finite set, and so we can write

$$I_{\sigma(M_g)} = \sum_{\lambda \in \text{ess im}(g)} \lambda \chi_{g^{-1}(\lambda)}$$

then

$$\begin{aligned}
\left(\int_{\sigma(M_g)} I_{\sigma(M_g)} dE \right) f &= \left(\int_{\sigma(M_g)} \sum_{\lambda \in \text{ess im}(g)} \lambda \chi_{g^{-1}(\lambda)} dE \right) f \\
&= \left(\sum_{\lambda \in \text{ess im}(g)} \lambda \int_{\sigma(M_g)} \chi_{g^{-1}(\lambda)} dE \right) f \\
&= \left(\sum_{\lambda \in \text{ess im}(g)} \lambda E(\{\lambda\}) \right) f \\
&= \left(\sum_{\lambda \in \text{ess im}(g)} \lambda M_{\chi_{g^{-1}(\lambda)}} \right) f \\
&= \left(\sum_{\lambda \in \text{ess im}(g)} \lambda \chi_{g^{-1}(\lambda)} \right) f
\end{aligned}$$

This last sum is equivalent to g almost everywhere. Therefore,

$$\begin{aligned}
\left(\int_{\sigma(M_g)} I_{\sigma(M_g)} dE \right) f &= gf \\
&= M_g f
\end{aligned}$$

As f was arbitrary, we have

$$M_g = \int_{\sigma(M_g)} I_{\sigma(M_g)} dE$$

as desired. Taking the limit of simple functions, we obtain the general case for the multiplication by an essentially bounded $g : \mathbb{R} \rightarrow \mathbb{C}$.

Example 3.13. As a final example, let's look at the shift operator on $L^2(\mathbb{R})$. Let $ST_a : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ be the shift operator

$$(T_a f)(t) = f(t - a)$$

In the Fourier transform example, we saw

$$\begin{aligned}
\mathcal{F}T_a(f(t)) &= \mathcal{F}(T_a f(t)) \\
&= \mathcal{F}f(t - a) \\
&= e^{-2\pi i \omega a} \mathcal{F}(f(t)) \\
&= M_{\exp(-2\pi i \omega a)} \mathcal{F}(f(t))
\end{aligned}$$

To simplify notation, write M_a for $M_{\exp(-2\pi i \omega a)}$. Then, we have $T_a = \mathcal{F}^* M_a \mathcal{F}$. The spectrum of M_a is the unit circle in the complex plane,

$$\sigma(M_a) = \{z \in \mathbb{C} \mid |z| = 1\}$$

Let $\lambda \in \sigma(M_a)$, then $M_a - \lambda I_{L^2(\mathbb{R})}$ is not invertible. Using the unitary equivalence of T_a and M_a , we have

$$\begin{aligned} M_a - \lambda I_{L^2(\mathbb{R})} \text{ not invertible} &\iff \mathcal{F}T_a\mathcal{F} - \lambda\mathcal{F}\mathcal{F}^* \text{ not invertible} \\ &\iff \mathcal{F}(T_a - \lambda I_{L^2(\mathbb{R})})\mathcal{F}^* \text{ not invertible} \end{aligned}$$

but $\mathcal{F}$ is invertible, and so we must have $T_a - \lambda I_{L^2(\mathbb{R})}$ is not invertible. Thus $\lambda \in \sigma(T_a)$. Then, we have $\sigma(T_a) = \sigma(M_a)$. With this in mind, define the map $K : \text{Borel}(\sigma(T_a)) \rightarrow \mathcal{B}(L^2(\mathbb{R}))$ by

$$K(S) = \mathcal{F}^*E(S)\mathcal{F}$$

where E is the projection-valued measure for M_a and $S \in \text{Borel}(\sigma(T_a))$. I claim that K is also a projection-valued measure. To see this, notice that for any $S \in \text{Borel}(\sigma(T_a))$, we have

$$\begin{aligned} K(S)^2 &= K(S)K(S) \\ &= (\mathcal{F}^*E(S)\mathcal{F})(\mathcal{F}^*E(S)\mathcal{F}) \\ &= \mathcal{F}^*E(S)\mathcal{F}\mathcal{F}^*E(S)\mathcal{F} \\ &= \mathcal{F}^*E(S)I_{L^2(\mathbb{R})}E(S)\mathcal{F} \\ &= \mathcal{F}^*E(S)E(S)\mathcal{F} \\ &= \mathcal{F}^*E(S)\mathcal{F} && (\text{As } E(S) \text{ is idempotent}) \\ &= K(S) \end{aligned}$$

and

$$\begin{aligned} K(S)^* &= (\mathcal{F}^*E(S)\mathcal{F})^* \\ &= \mathcal{F}^*E(S)^*\mathcal{F}^{**} \\ &= \mathcal{F}^*E(S)\mathcal{F} && (\text{As } E(S)^* = E(S) \text{ and } \mathcal{F}^{**} = \mathcal{F}) \\ &= K(S) \end{aligned}$$

Thus, $K(S)$ is idempotent and self-adjoint, hence is a projection. The other properties required for a projection-valued measure follow from the fact that E is a projection-valued measure.

4 The Functional Calculus for Bounded Normal Operators

Definition 4.1. Let $N \in \mathcal{B}(\mathcal{H})$ be a normal operator, and let E be the spectral measure for N . A positive measure μ is called a *scalar-valued spectral measure* (SVSM) for N if μ and E are absolutely continuous with respect to each other, i.e. for any measurable $S \in \text{Borel}(\sigma(N))$, $E(S) = 0$ if and only if $\mu(S) = 0$

The perspective behind the scalar-valued spectral measure is to allow us to connect functions that take on real-valued integrals with linear operators. The absolute continuity condition ensures that the measures agree on relevant sets.

In the buildup to the spectral theorem, we showed that the map $\phi \mapsto \int_{\sigma(N)} \phi dE$ is a homomorphism. In that section, we showed this for bounded functions. Here, under the functional calculus, as we will have a scalar-valued measure on $\sigma(N)$, we can extend what we did then to μ -essentially bounded functions. Moreover, doing this will give us some additional properties.

Definition 4.2. Let $T \in \mathcal{B}(\mathcal{H})$. A vector $f \in \mathcal{H}$ is called a cyclic vector for T if the set

$$\{T^k f \mid k \in \mathbb{N}_0\}$$

spans $\mathcal{H}$.

Example 4.3. As an example, consider the map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by the matrix

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

and consider the vector $\vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. The set

$$\begin{aligned} \{\vec{v}, T\vec{v}\} &= \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \\ &= \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \end{aligned}$$

spans all of $\mathbb{R}^2$. Thus, $\vec{v}$ is a cyclic vector for T . In fact, every nonzero vector $\vec{u} \in \mathbb{R}^2$ is a cyclic vector for T . More generally, if $\mathcal{A} \subseteq \mathcal{B}(\mathcal{H})$, then a vector $f \in \mathcal{H}$ is a cyclic vector for $\mathcal{A}$ if the set

$$\{T^k f \mid T \in \mathcal{A}, k \in \mathbb{N}_0\}$$

spans $\mathcal{H}$.

Definition 4.4. Let $\mathcal{A} \subseteq \mathcal{B}(\mathcal{H})$. A vector $f \in \mathcal{H}$ is called a separating vector for $\mathcal{A}$ if $Tf = 0 \implies T = 0$ for all $T \in \mathcal{A}$.

Definition 4.5. Let $\mathcal{A} \subseteq \mathcal{B}(\mathcal{H})$. The set

$$\{U \in \mathcal{B}(\mathcal{H}) \mid UT = TU \text{ for all } T \in \mathcal{A}\}$$

is called the commutant of $\mathcal{A}$, and is denoted $\mathcal{A}'$.

Remark 4.6. If $T \in \mathcal{A} \subseteq \mathcal{B}(\mathcal{H})$, then for all $U \in \mathcal{A}'$, we have $TU = UT$, i.e. $T \in \mathcal{A}''$. As T was arbitrary, we have that $\mathcal{A}$ is contained in its double commutant $\mathcal{A}''$.

Proposition 4.7. Let $\mathcal{A} \subseteq \mathcal{B}(\mathcal{H})$. If f is a cyclic vector for $\mathcal{A}$, then f is a separating vector for $\mathcal{A}'$.

Indeed, suppose that $Uf = 0$ for some $U \in \mathcal{A}'$. Then, for every $T \in \mathcal{A}$

$$UTf = TUF = 0$$

By taking powers of T , we have $T^k Uf = UT^k f = 0$ for all $k \in \mathbb{N}_0$. But remember that $\{T^k f \mid T \in \mathcal{A}, k \in \mathbb{N}_0\}$ spans $\mathcal{H}$. Thus, U sends a basis of $\mathcal{H}$ to 0. By linearity, we then have $U = 0$. Therefore, f is a separating vector for $\mathcal{A}'$.

Theorem 4.8. *The Functional Calculus for a Normal Operator. Let $\mathcal{H}$ be a separable Hilbert space, let $N \in \mathcal{B}(\mathcal{H})$ be a normal operator, and let μ be an SVSM for N . Then, the map $\mathfrak{U}: L^\infty(\mu) \rightarrow W^*(N)$ given by $\mathfrak{U}(f) = \int_{\sigma(N)} f dE$ is well-defined and*

(i) $\mathfrak{U}$ is a $*$ -isomorphism

(ii) $\mathfrak{U}: (L^\infty(\mu), \text{wk}^*) \rightarrow (W^*(N), \text{WOT})$ is a homeomorphism

Because of the definition of $\mathfrak{U}$, it is customary to write $f(N) = \mathfrak{U}(f)$.

Proof. In the construction of the spectral theorem, we defined a similar map for bounded functions on $\sigma(N)$. Now that we have a scalar measure, we can extend what we did there to apply to μ -essentially bounded functions. Let $\phi \in L^\infty(\mu)$ and suppose $\phi = 0$ a.e. $[\mu]$. As μ and E are mutually absolutely continuous, we have $\int_{\sigma(N)} \phi dE = 0$ a.e.. By linearity of the integral, we have that there is a unique $\mathfrak{U}(\phi)$ for every $\phi \in L^\infty(\mu)$, so $\mathfrak{U}$ is well-defined.

Let $\gamma = \sum_{k=1}^n a_k \chi_{S_k}$ and $\phi = \sum_{\ell=1}^m b_\ell \chi_{Q_\ell}$ be simple functions in $L^\infty(\mu)$. Then,

$$\begin{aligned}
 \mathfrak{U}(\phi\gamma) &= \int_{\sigma(N)} \phi\gamma dE \\
 &= \int_{\sigma(N)} \left(\sum_{k=1}^n a_k \chi_{S_k} \right) \left(\sum_{\ell=1}^m b_\ell \chi_{Q_\ell} \right) dE \\
 &= \int_{\sigma(N)} \left(\sum_{k=1}^n \sum_{\ell=1}^m a_k b_\ell \chi_{S_k} \chi_{Q_\ell} \right) dE \\
 &= \sum_{k=1}^n \sum_{\ell=1}^m \left(a_k b_\ell \int_{\sigma(N)} \chi_{S_k \cap Q_\ell} dE \right) \\
 &= \sum_{k=1}^n \sum_{\ell=1}^m a_k b_\ell E(S_k \cap Q_\ell) \\
 &= \sum_{k=1}^n \sum_{\ell=1}^m a_k b_\ell E(S_k) E(Q_\ell) \\
 &= \left(\sum_{k=1}^n a_k E(S_k) \right) \left(\sum_{\ell=1}^m b_\ell E(Q_\ell) \right) \\
 &= \left(\sum_{k=1}^n a_k \int_{\sigma(N)} \chi_{S_k} dE \right) \left(\sum_{\ell=1}^m b_\ell \int_{\sigma(N)} \chi_{Q_\ell} dE \right) \\
 &= \left(\int_{\sigma(N)} \sum_{k=1}^n a_k \chi_{S_k} dE \right) \left(\int_{\sigma(N)} \sum_{\ell=1}^m b_\ell \chi_{Q_\ell} dE \right) \\
 &= \left(\int_{\sigma(N)} \phi dE \right) \left(\int_{\sigma(N)} \gamma dE \right)
 \end{aligned}$$

Like before, by taking limits of simple functions, we get that $\mathfrak{U}$ is a homomorphism of C^* -algebras. Now, because E and μ are mutually absolutely continuous, we have $\mathfrak{U}$ is injective:

Let $\phi \in L^\infty(\mu)$ and suppose $\mathfrak{U}(|\phi|) = \int_{\sigma(N)} |\phi| dE = 0$. Then, because $|\phi| \geq 0$, $|\phi| = 0$ a.e. $[E]$, and hence 0 a.e. $[\mu]$. The proof of surjectivity is left out due to time and length constraints, but more information can be found in [2], [3]. Thus, $\mathfrak{U}$ is a *-isomorphism.

Now, let $f \in \mathcal{H}$ be a separating vector for $W^*(N)$ such that $\mu(S) = \langle E(S)f, f \rangle$. Let $\{\gamma_j\}_{j \in I}$ be a net in $L^\infty(\mu)$. Suppose $\mathfrak{U}(\gamma_j)$ converges to 0 under the WOT. Let $\phi \in L^1(\mu)$ and $\phi \geq 0$. Then the measure $\phi\mu$ is absolutely continuous with respect to μ . Then, there is a vector $h \in \mathcal{H}$ such that for any measurable $S \in \text{Borel}(\sigma(N))$, we have $\phi\mu(S) = \langle E(S)h, h \rangle = \mu_h(S)$. Then,

$$\begin{aligned} \langle \mathfrak{U}(\gamma_j)h, h \rangle &= \int_{\sigma(N)} \gamma_j d\mu_h \\ &= \int_{\sigma(N)} \gamma_j \phi d\mu \end{aligned}$$

As $\mathfrak{U}(\gamma_j)$ converges WOT to 0, we have that this integral converges to 0. But $\phi \in L^1(\mu)$ was arbitrary, and so we get γ_j converges wk^* to 0.

For the other direction, suppose γ_j converges wk^* to 0. and let $g, k \in \mathcal{H}$, then there is a function $\phi \in L^1(\mu)$ such that $\phi\mu = \mu_{g,k}$, and so

$$\begin{aligned} \langle \mathfrak{U}(\gamma_j)g, k \rangle &= \int_{\sigma(N)} \gamma_j d\mu_{g,k} \\ &= \int_{\sigma(N)} \gamma_j \phi d\mu \end{aligned}$$

As g, k were arbitrary, we conclude that $\mathfrak{U}(\gamma_j)$ converges WOT to 0. Therefore, the map $\mathfrak{U}$ is continuous, and has continuous inverse, hence is a homeomorphism. $\square$

5 Next Steps and Conclusion

The functional calculus allows us relate essentially bounded functions on $\sigma(N)$ with operators in $W^*(N)$ in a very structured way. There is a lot more we could do with normal operators, such as the spectral mapping theorem, which shows that for any function $\phi \in L^\infty(\mu)$, the spectrum of $\mathfrak{U}(\phi)$ is a.e. equivalent to the image of the spectrum of N , $\phi(\sigma(N))$.

Theorem 5.1. *The Spectral Mapping Theorem. Let $\mathcal{H}$ be a separable Hilbert space, let $N \in \mathcal{B}(\mathcal{H})$ be a normal operator, let μ be a scalar-valued spectral measure for N , and let $f \in L^\infty(\mu)$. Then, the spectrum of $f(N)$ is the μ -essential range of f*

Throughout this entire paper, we almost exclusively dealt with the theory of normal operators as it pertains to bounded operators. However, not all linear operators are bounded. The exploration of these operators is, unfortunately, too vast to fit into the margins of this paper. and so we will have to stop here for now. Nevertheless, this paper has covered a large amount of content, invoking results from topology and measure theory, to build a theorems relating to the spectra of normal operators, and the various spaces involved in these theorems.

A notion that was skipped however, is that the spectral theorem also provides a way in which to decompose the underlying space. In the finite dimensional case, this results in representing a vector space as a direct sum of the eigenspaces of a linear operator. In the

infinite dimensional case however, we've seen that elements of the spectrum of a normal operator need not be eigenvalues. The decomposition of the space takes on a slightly different meaning and methodology as a result.

Acknowledgements. This paper was written under the supervision of Dr. Ryan Hamilton at the University of Calgary.

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